

Ebad Rehman

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Education

University of Calgary – B.Sc. in Software Engineering

Graduation: May 2027

Certifications – AWS Solutions Architect - Associate

Jan 2026

Experience

Software Engineering Intern, TechQuest – Calgary, AB

June 2025 – August 2025

- Reduced manual workflow configuration time by 60% in multi-agent AI systems using LangChain and LangGraph, automating 5+ pipelines with n8n for autonomous task execution.
- Architected serverless AWS infrastructure (Lambda, S3, DynamoDB, CloudWatch) to process 1,000+ daily IoT events, reducing infrastructure costs by 30%.
- Delivered 3 production AI agents (weather forecasting, EV configuration, intelligent search) by collaborating with an 8-engineer team on system architecture reviews and API integrations.

Full Stack Developer, Freelance – Calgary, AB

December 2023 – Present

- Architected and deployed high-performance web platforms for commercial clients using Next.js (React) and TypeScript, implementing Static Site Generation (SSG) to maximize SEO ranking and load speeds.
- Engineered a custom routing system to support bi-directional text (English/Arabic), achieving 100% UI consistency across Right-to-Left (RTL) and LTR layouts for mobile and desktop viewports.
- Sustained 3,000+ monthly active users across client portfolios by integrating Google Analytics and Search Console data to drive continuous performance optimizations and content strategy.

Projects

FightMetrics — [AI-Powered Athletic Performance Tracker](#)

- Built a recommendation engine to optimize athlete training schedules by applying K-Means clustering (Scikit-Learn) to identify ineffective workout patterns and statistical distribution analysis to flag neglected disciplines.
- Engineered an automated injury-prevention system using Pandas rolling windows, enabling the platform to process 50+ daily data points per user and identify high-risk "Acute:Chronic" workload ratios without manual input.
- Deployed a real-time burnout prediction pipeline that aggregates 7-day vs. 28-day load averages, utilizing Socket.IO to push live injury-risk alerts to a React/TypeScript dashboard.

Real-Time Signal Processing System — [STM32 / ARM Cortex-M](#)

- Engineered a bare-metal embedded system on an STM32 (ARM Cortex-M4) using C++, bypassing standard HAL libraries to directly manipulate hardware registers for optimized I/O performance.
- Implemented a Zero-Copy data pipeline using Direct Memory Access (DMA) and Circular Buffers, allowing high-frequency ADC sampling (10+ kHz) with less than 5% CPU utilization.
- Designed a hardware interrupt-driven scheduler to manage concurrent tasks and timing-critical events, ensuring deterministic execution for real-time signal analysis.

LED Intensity Controller — [Embedded PIC24F Software](#)

- Engineered an end-to-end data pipeline streaming 10 samples/second from a PIC24F microcontroller ADC via UART, successfully feeding real-time potentiometer data from a mechanical knob to a Python visualization script.
- Built a real-time visualization dashboard using Matplotlib, which included automated CSV logging, enabling immediate trend monitoring and historical data analysis of LED intensity changes from a PWM knob coded in C.

Skills

Languages: Python, Java, C, C++, JavaScript, TypeScript, SQL, Bash (Linux)

Cloud & Infrastructure: AWS (Lambda, S3, DynamoDB, EC2), Serverless Architecture, ETL Pipelines

Frameworks: React, Next.js, Express.js, Flask, LangChain, LangGraph

Tools: Git, MySQL, NoSQL (MongoDB), n8n